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# Read the following rules carefully:

- Write your name and ID number in the solution you submit. ✓
- Please sign and submit the code of honor in the exam with your solutions. The exam cannot be graded without a signed code of honor. You are supposed to do all of the problems on your own without receiving help from others. ✓
- Cheating in the exam will not be tolerated. If you are caught cheating, you will be reported to the authorities. The recommendation of the instructor will be at least an F in the course in such cases. ✓
- Do not post any questions on Piazza about the exam. The TAs have been instructed to refrain from answering questions about the midterm. In case of ambiguity or problems in the questions, just do your best. ✓
- The use of generative AI is prohibited in answering the questions. ✓
- Problems are not sorted in terms of difficulty. Please avoid guess work and long and irrelevant answers. ✓
- Instructions on submitting the solutions to paper and pencil and coding problems will be provided shortly by the TAs. You can handwrite or typeset your solutions to paper and pencil problems. ✓
- Show all your work and your final answer. Showing only the final answer of a question may not receive full credit and you must show your solution and reasoning behind the answer. Simplify your answer as much as you can. ✓
- The exam has 8 questions, 16 pages, and a total of 100 points. ✓
- The submission deadline for this midterm is 11:59 PM, Saturday 24, 2025. ✓
- As this is a take home exam that extends over several days, OSAS students DO NOT receive extra time, per OSAS guidelines. ✓
- Any change in the midterm (paper and pencil or coding) after the deadline is considered late submission. One second late is late. The midterm is graded based on *when it was submitted, not when it was finished.* The midterm can be submitted up to two days late, with 10% penalty per late day. Homework late days cannot be used for the midterm. ✓
- Submission after the grace period will receive a zero. One second late is late. ✓

**Grading Breakdown**

| Problem | Score | Earned |
|---|---|---|
| 1 | 10 | |
| 2 | 10 | |
| 3 | 10 | |
| 4 | 10 | |
| 5 | 10 | |
| 6 | 10 | |
| 7 | 20 | |
| 8 | 20 | |
| Total | 100 | |

**Honor Code**

I pledge on my honor that I have not given or received any unauthorized assistance on this examination.

Name: Alyssa Keehan

Signature: 

1. Although in practice, we usually model conditional distributions of features in each class as Gauusians for Bayesian Discriminant Analysis, one must notice that Bayesian Discriminant Analysis is doable using any type of distribution. Assume that we wish to perform Discriminant Analysis with only one positive predictor $x \geq 0$ (i.e. $p = 1$), but instead of normal class pdfs, we have Lognormal class pdfs whose formula is:

$$f_k(x) = \frac{1}{x\sqrt{2\pi}\sigma_k} \exp\left(\frac{-(\ln x - \mu_k)^2}{2\sigma_k^2}\right), x \geq 0$$

We know that the posterior class probability for class $k$ is:

$$\Pr(Y = k|X = x) = p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

where $\pi_l$'s are prior class probabilities.

(a) For $K = 3$ and $\pi_1 = \pi_2 = \pi_3$ and $\sigma_1 = 1, \sigma_2 = 2, \sigma_3 = 3$ and $\mu_1 = 1, \mu_2 = 5, \mu_3 = 7$, determine what ranges of $x$ are classified in each class. Hint: plotting the pdfs may help.

(b) For binay classification ($K = 2$), when classes are balanced ($\pi_1 = \pi_2$), show that when $\sigma_1 = \sigma_2$, the decision boundary between the two classes is the threshold

$$x^* = \sqrt{\exp(\mu_1)}\sqrt{\exp(\mu_2)}$$

a) priors are equal: $\pi_1 = \pi_2 = \pi_3$ → same for every class

take log → $f_k(x) = -\ln(x) - \frac{1}{2}\ln(2\pi) - \ln(\sigma_k) - \frac{(\ln(x) - \mu_k)^2}{2\sigma_k^2}$

let $t = \ln(x)$

$= -\ln(\sigma_k) - \frac{(t-\mu_k)^2}{2\sigma_k^2}$

class 1 ($\sigma_1 = 1, \mu_1 = 1$) $= \ln(1) - \frac{(t-1)^2}{2(1)^2} = -\frac{(t-1)^2}{2}$

class 2 ($\sigma_2 = 2, \mu_2 = 5$) $= \ln(2) - \frac{(t-5)^2}{2(2)^2} = \ln(2) - \frac{(t-5)^2}{8}$

class 3 ($\sigma_3 = 3, \mu_3 = 7$) $= \ln(3) - \frac{(t-7)^2}{2(3^2)} = \ln(3) - \frac{(t-7)^2}{18}$

Class 1 vs. Class 2:

$$\frac{-(t-1)^2}{2} = -\ln(2) - \frac{(t-5)^2}{8}$$

$$4(t-1)^2 = 8\ln(2) + (t-5)^2$$

$$4t^2 - 8t + 4 = t^2 - 10t + 25 + 8\ln(2)$$

$$3t^2 + 2t - 21 - 8\ln(2) = 0$$

$$3t^2 + 2t - 26.545 = 0$$

$$t = \frac{-(2) \pm \sqrt{2^2 - 4(3)(-26.545)}}{2(3)} = \frac{-2 \pm \sqrt{4 + 318.54}}{6} = \frac{-2 \pm \sqrt{322.54}}{6}$$

$$= \frac{-2 \pm 17.959}{6} \Rightarrow t = 2.66 \text{ or } t = -3.33$$

Class 1 vs Class 3

$$\frac{-(t-1)^2}{2} = -\ln(3) - \frac{(t-7)^2}{18}$$

$$-9(t-1)^2 = -18\ln(3) - (t-7)^2$$

$$0 = 9(t-1)^2 - (t-7)^2 - 18\ln(3)$$

$$= 9t^2 - 18t^2 + 9 - t^2 + 14t - 49 - 18\ln(3)$$

$$= 8t^2 - 4t - 40 - 18\ln(3)$$

$$= 8t^2 - 4t - 40 - 19.775$$

$$= 8t^2 - 4t - 59.775$$

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(8)(-59.775)}}{2(8)}$$

$$= \frac{4 \pm \sqrt{16 + 1,912.80}}{16} = \frac{4 \pm \sqrt{1928.80}}{16}$$

$$= \frac{4 \pm 43.918}{16} \quad t = 2.995 \text{ or } t = -2.495$$

Class 2 vs. Class 3

$$-\ln(2) - \frac{(t-5)^2}{8} = -\ln(3) - \frac{(t-7)^2}{18}$$

$$-72\ln(2) - 9(t-5)^2 = -72\ln(3) - 4(t-7)^2$$

$$-9t^2 + 90t - 225 - 72\ln(3) = -4t^2 + 56t - 196 - 72\ln(3)$$

$$-5t^2 + 34t - 29 = 72\ln 3 - 72\ln(2)$$

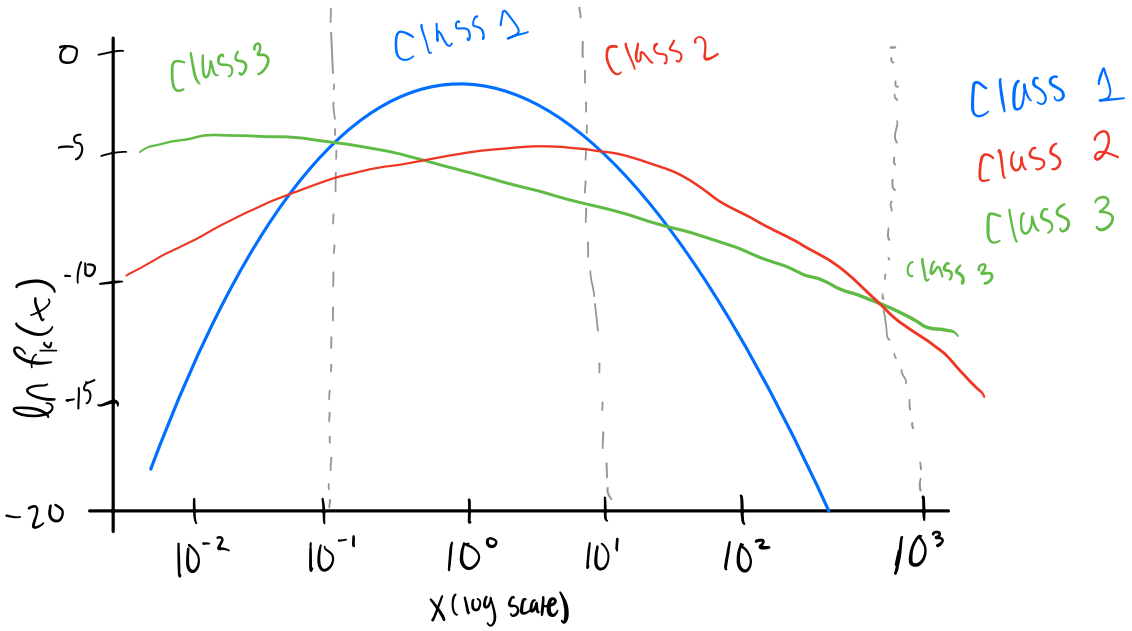
$$-5t^2 + 34t - 29 = 79.10 - 49.91$$

$$-5t^2 + 34t - 29 + 29.19 = 0$$

$$-5t^2 + 34t + 0.19 = 0$$

$$t = \frac{-34 \pm \sqrt{34^2 - 4(-5)(0.19)}}{2(-5)} = \frac{34 \pm \sqrt{1,156 + 3.8}}{10}$$

$$= \frac{34 \pm 34.056}{10} \quad t = 6.81 \text{ or } t = -0.006$$




winner changes from Class 3 → class 1 @ t = -2.495

Class 1 → class 2 @ t = 2.66

Class 2 → class 3 @ t = 6.806

$t = \ln(x)$

$x = e^t$

class 1: $e^{-2.495} = 0.0825$

class 2: $e^{2.660} = 14.296$

class 3: $e^{6.806} = 903.251$

$0 < x < 0.0825 \rightarrow$ class 3

$0.0825 < x < 14.296 \rightarrow$ class 1

$14.296 < x < 903.251 \rightarrow$ class 2

$x > 903.251 \rightarrow$ class 3

b) $\exp\left\{\frac{-(\ln(x)-\mu_1)^2}{2\sigma^2}\right\} = \exp\left\{\frac{-(\ln(x)-\mu_2)^2}{2\sigma^2}\right\}$

$$\frac{-(\ln(x)-\mu_1)^2}{2\sigma^2} = \frac{-(\ln(x)-\mu_2)^2}{2\sigma^2}$$

$$-\ln(x)^2 + 2\mu_1\ln(x) - \mu_1^2 = -\ln(x)^2 + 2\mu_2\ln(x) - \mu_2^2$$

$$2\mu_1\ln(x) - 2\mu_2\ln(x) = \mu_1^2 - \mu_2^2$$

$$2\ln(x)(\mu_1 - \mu_2) = (\mu_1 - \mu_2)(\mu_1 + \mu_2)$$

$$2\ln(x) = \mu_1 + \mu_2$$

$$\ln(x) = \frac{\mu_1 + \mu_2}{2}$$

$$e \qquad e$$

$$x = e^{\frac{\mu_1+\mu_2}{2}} = (e^{\mu_1}e^{\mu_2})^{1/2} = \sqrt{e^{\mu_1}}\sqrt{e^{\mu_2}}$$

$$\therefore x^* = \sqrt{\exp(\mu_1)}\sqrt{\exp(\mu_2)}$$

2. Consider the following unusual logistic regression model for binary classification:

$$\Pr(Y=1|X_1,X_2)=\frac{\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}{1+\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}$$

we wish to fit the model to the following dataset:

| Index | $X_1$ | $X_2$ | $Y$ |
|---|---|---|---|
| 1 | 1 | -1 | 1 |
| 2 | 1 | 1 | 1 |
| 3 | -1 | 1 | 0 |
| 4 | -1 | -1 | 0 |

(a) Write down the log odds model for this unusual logistic regression model.

(b) Write down the *likelihood function* for this dataset.

(c) Analytically find the parameters $\beta_i$ that maximize the likelihood function (or minimize the negative log likelihood function.)

a) let $z=\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}$

$\Pr(y=1|X_1,X_2)=\frac{z}{1+z} \Rightarrow 1-p=1-\frac{z}{1+z}=\frac{1+z-z}{1+z}=\frac{1}{1+z}$

odds: $\frac{p}{1-p}=\frac{\frac{z}{1+z}}{\frac{1}{1+z}}=\frac{z}{1+z}\cdot\frac{1+z}{1}=z=\beta_0|X_1|^{\beta}|X_2|^{\beta_2}$

log odds: $\ln\left(\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}\right)=\ln(\beta_0)+\beta_1\ln(|X_1|)+\beta_2\ln(|X_2|)$

b)

| row | $\lvert X_1\rvert^{\beta_1}\lvert X_2\rvert^{\beta_2}$ | y |
|---|---|---|
| 1 | $1^{\beta_1}\cdot 1^{\beta_2}=1$ | 1 |
| 2 | $1^{\beta_1}\cdot 1^{\beta_2}=1$ | 1 |
| 3 | $1^{\beta_1}\cdot 1^{\beta_2}=1$ | 0 |
| 4 | $1^{\beta_1}\cdot 1^{\beta_2}=1$ | 0 |

⇒ because 1 to the power of anything, $\beta_1$ & $\beta_2$ become obsolete.

NOW → $p=\frac{\beta_0}{1+\beta_0}$ & $1-p=\frac{1}{1+\beta_0}$

Likelihood: $L=\prod_{i:y=1}^{4}p(x_i)\prod_{i:y=0}^{4}(1-p(x_i))=\frac{\beta_0}{1+\beta_0}\cdot\frac{\beta_0}{1+\beta_0}\cdot\frac{1}{1+\beta_0}\cdot\frac{1}{1+\beta_0}$

$\therefore$ Likelihood $=\frac{\beta_0^2}{(1+\beta_0)^4}$

c) -log likelihood: $= -\left[\ln\left(\frac{\beta_0^2}{(1+\beta_0)^4}\right)\right] = -\left[\ln(\beta_0^2) - \ln((1+\beta_0)^4)\right]$

$= -2\ln(\beta_0) + 4\ln(1+\beta_0)$

take derivative → $\frac{d\ell}{d\beta_0} = -2 \cdot \frac{1}{\beta_0} + 4 \cdot \frac{1}{1+\beta_0} = -\frac{2}{\beta_0} + \frac{4}{1+\beta_0}$

$\frac{2}{\beta_0} = \frac{4}{1+\beta_0} \Rightarrow 2 + 2\beta_0 = 4\beta_0$

$\frac{2}{2} = \frac{2\beta_0}{2}$

$\beta_0 = 1$ → $\hat{\beta_0} = 1$ maximizes likelihood function

take 2nd derivative: $\frac{d^2\ell}{d\beta_0^2} = \frac{2}{\beta_0^2} - \frac{4}{(1+\beta_0)^2}$

@ $\beta_0 = 1$ : $\frac{2}{1} - \frac{4}{2^2} = 2 - 1 = 1$

Because $\beta_1$ & $\beta_2$ are cancelled out in likelihood & derivative of either $= 0$, we cannot identify parameters for those coefficients that maximize $\beta_1$ or $\beta_2$.

3. The regularized least squares problem has the following objective function:

$$J(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda R(\beta_0, \beta_1, \ldots, \beta_p)$$
$$= \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \cdots - \beta_p x_{ip})^2 + \lambda R(\beta_0, \beta_1, \ldots, \beta_p)$$

Determine which of the following choices of $R(\beta_0, \beta_1, \ldots, \beta_p)$ are *valid* regularizers, in the sense of shrinking the parameters, and explain your answer. Also explain among the valid regularizers, which of them tends to yield sparse solutions for $\beta_i$'s. Hint: for simplicity, you can think about each of them when applied to a regression problem with only one regressor.

(a) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \beta_i^3$

(b) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \ln(1 + |\beta_i|)$

(c) $R(\beta_0, \beta_1, \ldots, \beta_p) = \|\beta\|_0)$, where the $\mathscr{L}_0$ norm of the vector $\beta$ is defined as the number of non-zero elements in $\beta = (\beta_0, \beta_1, \ldots, \beta_p)$. For example, if $\beta = (1, 2, 3, 4)$, then $\|\beta\|_0) = 4$ and for $\beta = (1000, 240, 0.03, 0)$, $\|\beta\|_0 = 3$, and for $\beta = (10^{-6}, 0.2, 0, 0)$, $\|\beta\|_0 = 2$.

(d) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \sin(\beta_i)$

(e) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} |\beta_i|$

(f) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \beta_i^4$

* penalty is smallest when $\beta_i = 0$; however cannot be negative.

a) not valid
- if $\beta < 0$, then $\beta_i^3$ will be negative & has no lower bound (converges to $-\infty$ instead of 0)

b) valid & sparse
- it will always be $\geq 0$ & 0 only @ $\beta = 0$

c) valid & sparse
- it will always be $\geq 0$ because it is essentially a count, so no decimals

d) not valid
- sine is periodic & can be negative, smallest isn't @ 0 it's @ $\beta = -\pi/2$

e) valid; sparse
- always $\geq 0$ because of absolute value; smallest @ $\beta = 0$

f) valid
- always $\geq 0$ because of even exponent

4. Consider the following multiclass dataset:

| Index | $X_1$ | $X_2$ | $Y$ |
|---|---|---|---|
| 1 | 12 | -1 | + |
| 2 | -3 | 0 | - |
| 3 | 2 | 4 | - |
| 4 | 12 | 1 | * |
| 5 | -5 | 17 | - |
| 6 | 11 | 8 | + |
| 7 | 13 | 10 | + |
| 8 | -6 | -1.5 | * |
| 9 | 21 | -6 | + |

For this dataset, determine the leave-one-out cross validation estimate of the misclassification error of a simple classifier that *classifies to the prior*, i.e always classifies to the most common class in its training set. If you ever encounter a tie, break it in favor of * and then +.

+: 1, 6, 7, 9 → 4
-: 2, 3, 5 → 3
*: 4, 8 → 2

1) Leave out a +:
+ → 3
- → 3
* → 2
} in a tie go to "+"
+ → 4 → every left out + is predicted correctly
- → 3
* → 2
4 correct

2) Leave out a -
+ → 4 → most common class so predicts +, but true class is -
- → 2
* → 2
3 incorrect

3) Leave out a *
+ → 4 → most common again, predicts +, but true class is *
- → 2
* → 1
2 incorrect

# 5 misclassified observations

LOOCV Error = $\frac{5}{9} \approx 0.556$

LOOCV misclassification Error ≈ 55.6%.

5. Consider the task of identifying alligators and crocodiles in a pond based on their respective lengths. The average length of an adult crocodile is about 15 feet, while the average length of an adult alligator is about 12 feet. Assuming that their length $X$ follows a Gaussian distribution with a standard deviation $\sigma$ equal to 2 feet: (20 pts)

   (a) Write down the conditional pdf of $X$ given being a crocodile.

   (b) Write down the conditional pdf of $X$ given being an alligator.

   (c) Assuming that their prior probabilities are the same, the ideal decision boundary (threshold for $X$ that distinguishes between two classes) is located at some length $x^*$ such that $P(\text{Crocodile}|X = x^*) = P(\text{Alligator}|X = x^*)$.

   (d) Repeat 5c when we know that the population of the crocs is twice as much the population of alligators in the pond.

   (e) Repeat 5c when we know that the population of the alligators is twice as much the population of crocs in the pond.

   (f) Repeat 5c when the prior probabilities are the same, but standard deviations are different; i.e., $\sigma(\text{Crocodile}) = 4$ and $\sigma(\text{Alligator}) = 2$.

$\mu_c = 15$

$\mu_a = 12$

$\sigma = 2$

$N(\mu, \sigma^2)$

a) $f(x \mid \text{crocodile}) = \frac{1}{\sqrt{2\pi}\cdot 2} \exp\left\{-\frac{1}{2(2^2)}(x-15)^2\right\}$

$p(x \mid \text{crocodile}) = \frac{\exp\left\{-\frac{(x-15)^2}{8}\right\}}{2\sqrt{2\pi}}$

b) $p(x \mid \text{alligator}) = \frac{\exp\left\{-\frac{(x-12)^2}{8}\right\}}{2\sqrt{2\pi}}$

c) $\frac{\exp\left\{\frac{-(x-15)^2}{8}\right\}}{2\sqrt{2\pi}} = \frac{\exp\left\{\frac{-(x-12)^2}{8}\right\}}{2\sqrt{2\pi}}$

$\ln\left[\exp\left\{\frac{-(x-15)^2}{8}\right\}\right] = \ln\left[\exp\left\{\frac{-(x-12)^2}{8}\right\}\right]$

$\frac{-(x-15)^2}{8} = \frac{-(x-12)^2}{8}$

$x^2 - 30x + 225 = x^2 - 24x + 144$

$81 = 6x$

$x^* = 13.5$

The ideal decision boundary exists @ $x^* = 13.5$ (exactly halfway between 12 & 15). If $x < 13.5$ then predict alligator, if $x > 13.5$ then predict crocodile.

d) $p(\text{croc}) = \frac{2}{3}$

$p(\text{alligator}) = \frac{1}{3} \Rightarrow \frac{2}{3}\left[\frac{\exp\left\{\frac{-(x-15)^2}{8}\right\}}{2\sqrt{2\pi}}\right] = \left[\frac{\exp\left\{\frac{-(x-12)^2}{8}\right\}}{2\sqrt{2\pi}}\right] \cdot \frac{1}{3}$

$\frac{\exp\left\{\frac{-(x-15)^2}{8}\right\}}{3\sqrt{2\pi}} = \frac{\exp\left\{\frac{-(x-12)^2}{8}\right\}}{2 \cdot 3\sqrt{2\pi}}$

$\ln\left[2\exp\left\{\frac{-(x-15)^2}{8}\right\}\right] = \left[\exp\left\{\frac{-(x-12)^2}{8}\right\}\right] \ln$

$\ln(2) - \frac{(x-15)^2}{8} = \frac{-(x-12)^2}{8}$

$\ln(2) = \frac{(x-15)^2 - (x-12)^2}{8}$

$8\ln(2) = x^2 - 30x + 225 - x^2 + 24 - 144$

$8\ln(2) = -6x + 81$

$8\ln(2) - 81 = -6x$

$x^* = 13.5 - \frac{4}{3}\ln(2) \approx 12.58$

The ideal decision boundary exists @ $x^* = 12.58$. The threshold moves down closer to alligator mean indicating a higher density of predicting crocodile when $x^* > 12.58$ vs. $x^* < 12.58$ when predicting alligator.

e) $p(\text{alligator}) = \frac{2}{3}$

$p(\text{crocodile}) = \frac{1}{3}$

$\Rightarrow \ln\left[\exp\left\{\frac{-(x-15)^2}{8}\right\}\right] = \left[2\exp\left\{\frac{-(x-12)^2}{8}\right\}\right]\ln$

$$\frac{-(x-15)^2}{8} = \ln(2) - \frac{(x-12)^2}{8}$$

$$\frac{(x-12)^2-(x-15)^2}{8} = \ln(2)$$

$$x^2 - 24x + 144 - x^2 + 30x - 225 = 8\ln(2)$$

$$6x - 81 = 8\ln(2)$$

$$6x = 8\ln(2) + 81$$

$$x^* = 13.5 + \frac{4}{3}\ln(2) \approx 14.42$$

The ideal decision boundary exists @ $x^* = 14.42$
The threshold moves up closer to crocodile mean indicating a higher density of predicting alligator when $x^* < 14.42$ vs. $x^* > 14.42$ when predicting crocodile.

f) $$\frac{\exp\left\{\frac{-(x-15)^2}{2\cdot 4^2}\right\}}{2\sqrt{2\pi}} = \frac{\exp\left\{\frac{-(x-12)^2}{8}\right\}}{2\sqrt{2\pi}}$$

$$\ln\left[\frac{1}{2}\cdot\exp\left\{\frac{-(x-15)^2}{32}\right\}\right] = \left[\exp\left\{\frac{-(x-12)^2}{8}\right\}\right]\ln$$

$$\ln\left[\ln(2) - \frac{(x-15)^2}{32}\right] = \left[\frac{-(x-12)^2}{8}\right]\cdot 32$$

$$32\ln(2) - (x-15)^2 = -4(x-12)^2$$

$$32\ln(2) = -4(x-12)^2 + (x-15)^2$$

$$32\ln(2) = -4x^2 + 96x - 576 + x^2 - 30x + 225$$

$$32\ln(2) = -3x^2 + 66x - 351$$

$$0 = -3x^2 + 66x - 351 - 32\ln(2)$$

$$0 = 3x^2 - 66x + 351 + 32\ln(2)$$

$$0 = 3x^2 - 66x + 351 + 27.18$$

$$0 = (3x^2 - 66x + 373.18)\frac{1}{3}$$

$$0 = x^2 - 22x + 124.39$$

$$x = \frac{-(-22) \pm \sqrt{22^2 - 4(1)(124.39)}}{2} = \frac{22 \pm \sqrt{484 - 497.56}}{2}$$

$$= \frac{22 \pm \sqrt{-13.56}}{2} \Rightarrow$$ because the discriminant is negative, there is no decision boundary

6. The following least squares liner regression model was fitted to a sample of 28 students using data obtained at the end of their senior year in college. The aim was to explain students' weight gains:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$
$$\hat{y} = 6.15 + 0.553 x_1 - 1.141 x_2 + 0.813 x_3$$

The standard errors are:

$$\text{SE}(\hat{\beta}_1) = 0.165$$
$$\text{SE}(\hat{\beta}_2) = 0.365$$
$$\text{SE}(\hat{\beta}_3) = 0.202$$

- $\hat{y}$: weight gained, in pounds, during senior year
- $x_1$: average number of meals eaten per week
- $x_2$: average number of hours of exercise per week
- $x_3$: average number of beers consumed per week

(a) Interpret the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$

(b) Test, at the 1% level, the null hypothesis that the true coefficient on the variable $x_3$ is 0 against the alternative that it is not 0.

(c) Find and interpret a 94% confidence interval for the parameter $\beta_1$.

(d) If for the model, SSR=83.4 (Regression Sum of Squares) and SSE = 43.9 (Residual Sum of Squares), test the hypothesis that all the coefficients of the model are 0 (test overall significance of the model) using $\alpha = 3\%$. .

→ same as RSS

a) $\hat{\beta_0} = 6.15$ (y-intercept)

6.15 lbs is the predicted weight gained for a student that eats on avg. 0 meals/week, exercises on avg. 0 hours a week & consumes on avg. 0 beers a week.

$\hat{\beta_1} = 0.553$

Holding avg # of hours of exercise & beers consumed/week constant, for every 1 meal eaten per week we expect a student to gain an extra 0.553 lbs during senior year

b) $H_0: \beta_3 = 0 \quad H_A: \beta_3 \neq 0$

under $H_0$: $t = \frac{\hat{\beta}_3 - \beta}{SE(\hat{\beta}_3)} = \frac{0.813 - 0}{0.202} \approx 4.02$

$df = n - p - 1 = 28 - 4 = 24$

with $t_{0.005,24} = 2.797 < 4.02$, we reject $H_0$. There is significant evidence that beer consumption has an effect on weight gain when holding meals & exercising constant.

c) for 94% confidence interval, we need $t_{0.03,24} \approx 1.974$

confidence interval: $\hat{\beta}_1 \pm 1.974\, SE(\hat{\beta}_1) = [0.553 \pm (1.974)(0.165)]$

$[0.553 \pm 0.32571] = [0.22729, 0.87871]$

We are 94% confident that the true value of $\beta_1$, holding all other variables constant falls between [0.227, 0.879]. Each extra meal/week adds anywhere between 0.227 and 0.879 lbs of weight gain during senior year.

d) $H_0: \beta_i = 0$ (no linear relationship vs. $H_A$: at least on $\beta_i \neq 0$

$RSS = 43.9$

Regression SS = 83.4

TSS = 127.3

$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$

$= \frac{(127.3 - 43.9)/3}{43.9/(28-3-1)} = \frac{83.4/3}{43.9/24} = \frac{27.8}{1.83} \approx 15.19 > f_{\alpha=0.03;3,24} = 3.53$

With f-statistic $\approx 15.19 > 3.53$, we reject $H_0$. There is significant evidence that at least one of the predictors explains the variability in weight gain during senior year.

7. The conditional probability distribution function of the weight $W$ (in kg) given the height $H$ (in cm) in a population is Gaussian (normal), and (35 pts)

$$p_{W|H}(w|h) = \frac{1}{\sqrt{2\pi} \times 10} \exp\left(-\frac{(w - 0.5 \times h^{1.001})^2}{200}\right)$$

E[W|H=h]

(a) What is the best estimate of the weight of a person as a function of their height in the sense of *mean squared error*? (Hint: look up the Gaussian distribution, its mean, and variance. This problem is asking you to determine the regression function $w = f(h)$, which was shown in the lecture to be a statistical property of the conditional distribution of the output $W$ given a particular value $h$ of the input $H$.)

(b) Use the result you obtained in 7a to estimate the weight of people whose heights are 155, 165, and 190 cm.

(c) Would your answer to 7a and 7b change if the conditional variance of $W$ given $H = h$ were a function of $h$, say, $\sigma(h)$, i.e.:

Same

$$p_{W|H}(w|h) = \frac{1}{\sqrt{2\pi} \times \sigma(h)} \exp\left(-\frac{(w - 0.5 \times h^{1.001})^2}{2[\sigma(h)]^2}\right)$$

(d) Assume that instead of the conditional distribution, you have the following sample. Estimate the weight of people whose heights are 150, 155, 165, and 190 cm, using KNN with $k = 3$:

$$\hat{y}_{KNN} = \frac{y_1 + y_2 + \cdots + y_k}{k}$$

where $y_1, y_2, \cdots, y_k$ are the labels of the $k$ nearest neighbors to your test instance.

| Person | Height (cm) | Weight (kg) |
|---|---|---|
| 1 | 171 | 80 |
| 2 | 168 | 78 |
| 3 | 191 | 100 |
| 4 | 182 | 80 |
| 5 | 150 | 65 |
| 6 | 178 | 83 |

(e) Repeat 7d, but instead of using the simple average of the labels of $k$ nearest neighbors, which is use the following weighted average:

$$\hat{y}_{KNN} = \frac{w_1 y_1 + w_2 y_2 + \cdots + w_k y_k}{w_1 + w_2 + \cdots + w_k}$$

where the weight $w_i$ for the label $y_i$ of instance $i$ is determined as $1/d_i$, where $d_i$ the distance between the instance $i$ and the test instance. **It is worth noting**

**that normalized weights can be viewed as similarities of the training instances with the test instance. An alternative formula would be:**

$$\hat{y}_{KNN} = s_1 y_1 + s_2 y_2 + \cdots + s_k y_k$$

where $s_i = sim(x_i, x^*)$, where $x^*$ is the test point.

a) $\mu = 0.5\,h^{1.001}$, $\sigma = 10$ } Normal Distribution

Under MSE, best estimate for conditional mean: $E[W|H=h] = 0.5\,h^{1.001}$

b)
$E[W|h=155] = 0.5(155)^{1.001} = 77.89$
$E[W|h=165] = 0.5(165)^{1.001} = 82.92$
$E[W|h=190] = 0.5(190)^{1.001} = 95.50$

c) NO, my answer for part a would not change. The best estimate only depends on conditional mean which is still $0.5\,h^{1.001}$.

d)
height = 150
(0) 150 → 65
(18) 168 → 78
(21) 171 → 80
$\frac{65+78+80}{3} = 74.33$

height = 155
(3) 150 → 65
(13) 168 → 78
(14) 171 → 80
$\approx 74.33$

height = 165
(3) 168 → 78
(6) 171 → 80
(17) 178 → 83
$\frac{78+80+83}{3} \approx 80.33$

height = 190

(1) 191 → 100

(8) 182 → 80

(12) 178 → 83

$\frac{100+80+83}{3} = 87.67$

e) height = 150:

↳ because one neighbor has a distance of 0, weight is infinite.

estimate is that neighbor's weight = 65 kg.

height = 155:

$\text{weights} = \frac{1}{5} + \frac{1}{13} + \frac{1}{16} = 0.2 + 0.0769 + 0.0625 = 0.3394$

$\frac{65}{5} + \frac{78}{13} + \frac{80}{16} = 13 + 6 + 5 = 24$

$\frac{24}{0.3394} = 70.71 \text{ kg}$

height = 165

$\text{weights} = \frac{1}{3} + \frac{1}{6} + \frac{1}{13} = 0.333 + 0.167 + 0.0769$

$\frac{78}{3} + \frac{80}{6} + \frac{83}{13} = 26 + 13.33 + 6.385 = 45.715$

$\frac{45.715}{0.5769} = 79.24 \text{ kg}$

height = 190

$\text{weights} = \frac{1}{1} + \frac{1}{8} + \frac{1}{12} = 1 + 0.125 + 0.083 = 1.2083$

$\frac{100}{1} + \frac{80}{8} + \frac{83}{12} = 100 + 10 + 6.9167 = 116.9167$

$\frac{116.9167}{1.2083} = 96.76 \text{ kg}$

8. **Programming Question: Predicting Housing Prices with Linear Regression**

   **Dataset:** `student-mat.csv`

   **Link:** https://archive.ics.uci.edu/dataset/320/student+performance

   In this problem, you will build a linear regression model to predict housing prices in the USA based on various features. You will use Python and `scikit-learn` for this task.

   (a) **Data Exploration and Pre-processing**

   i. Import the `student-mat.csv` dataset as a pandas DataFrame.

   ii. Select the following features: `age, studytime, schoolsup, goout, Dalc, Walc, health, absences, G3` (target/dependent variable). Encode the binary variable (`schoolsup`) values as 0s (no) and 1s (yes). Combine `Dalc` and `Walc` into one variable `alc` by taking averge, then remove `Dalc` and `Walc`. Display the first five rows of the pre-processed dataset.

   iii. Find the number of outliers for each independent variable using the IQR method.

   iv. Standardize and run PCA on the dataset. Create a scatterplot of PC1 vs PC2, coloring the dots by their final grade `G3`. Inspect the component loadings and determine which features contribute the most to PC1 and PC2. *Keep and use standardized data for remaining problems.*

   (b) **Linear Regression**

   i. Split data into training set and testing set with an 80:20 ratio. Use random seed 552 for reproducibility.

   ii. Build three models using the training set: A. Linear Regression Model, B. Linear Regression Model with Ridge Regularization, and C. Linear Regression Model with Lasso Regularization. Set $\alpha$=0.1.

   iii. Test all three models on the test set. Find out the best performing model with repsect to each of the metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and $R^2$.

   iv. How do you interpret $R^2$ values from the three models?

   v. Print coefficients of independent variables from the three models in one table.

   vi. What's the relationship between each independent variable and the dependent variable?

   vii. How do the regularization methods differ? What can you conclude about the dataset and features given the results?

   viii. If the regularization strength ($\alpha$) is increased, what would happen to performance metrics?

   ix. What are some feature engineering methods you would suggest to improve model performance?

   **Expected Output:** Your submission should include:

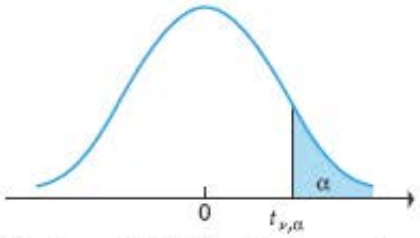
- Jupyter Notebook `.ipynb` with all the steps clearly commented.
- The output of each step as specified above (e.g., head of DataFrame, info, describe, missing value counts, evaluation metrics for all models, coefficients, and intercept).
- Visualizations for outlier detection and residual analysis.
- A brief discussion answering the interpretation questions.

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

| z | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| z | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

## Upper Critical Values of Student's $t$ Distribution with $\nu$ Degrees of Freedom




For selected probabilities, $\alpha$, the table shows the values $t_{\nu,\alpha}$ such that $P(t_\nu > t_{\nu,\alpha}) = \alpha$, where $t_\nu$ is a Student's $t$ random variable with $\nu$ degress of freedom. For example, the probability is .10 that a Student's $t$ random variable with 10 degrees of freedom exceeds 1.372.

| | Probability of Exceeding the Critical Value | | | | | |
|---|---|---|---|---|---|---|
| $\nu$ | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 | 0.001 |
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 318.313 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 22.327 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 10.215 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 7.173 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5.893 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 5.208 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.782 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 4.499 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 4.296 |
| 10 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 4.143 |
| 11 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 4.024 |
| 12 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.929 |
| 13 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.852 |
| 14 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.787 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.733 |
| 16 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.686 |
| 17 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.646 |
| 18 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 |
| 19 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 |
| 20 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 |
| 21 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 |
| 22 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.505 |
| 23 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.485 |
| 24 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.467 |
| 25 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.450 |
| 26 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.435 |
| 27 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.421 |
| 28 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.408 |
| 29 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.396 |
| 30 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.385 |
| 40 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 3.307 |
| 60 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 3.232 |
| 100 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 3.174 |
| $\infty$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 |
| $\nu$ | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 | 0.001 |

## F - Distribution ($\alpha = 0.01$ in the Right Tail)

| Denominator Degrees of Freedom $df_2$ \ Numerator Degrees of Freedom $df_1$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4052.2 | 4999.5 | 5403.4 | 5624.6 | 5763.6 | 5859.0 | 5928.4 | 5981.1 | 6022.5 |
| 2 | 98.503 | 99.000 | 99.166 | 99.249 | 99.299 | 99.333 | 99.356 | 99.374 | 99.388 |
| 3 | 34.116 | 30.817 | 29.457 | 28.710 | 28.237 | 27.911 | 27.672 | 27.489 | 27.345 |
| 4 | 21.198 | 18.000 | 16.694 | 15.977 | 15.522 | 15.207 | 14.976 | 14.799 | 14.659 |
| 5 | 16.258 | 13.274 | 12.060 | 11.392 | 10.967 | 10.672 | 10.456 | 10.289 | 10.158 |
| 6 | 13.745 | 10.925 | 9.7795 | 9.1483 | 8.7459 | 8.4661 | 8.2600 | 8.1017 | 7.9761 |
| 7 | 12.246 | 9.5466 | 8.4513 | 7.8466 | 7.4604 | 7.1914 | 6.9928 | 6.8400 | 6.7188 |
| 8 | 11.259 | 8.6491 | 7.5910 | 7.0061 | 6.6318 | 6.3707 | 6.1776 | 6.0289 | 5.9106 |
| 9 | 10.561 | 8.0215 | 6.9919 | 6.4221 | 6.0569 | 5.8018 | 5.6129 | 5.4671 | 5.3511 |
| 10 | 10.044 | 7.5594 | 6.5523 | 5.9943 | 5.6363 | 5.3858 | 5.2001 | 5.0567 | 4.9424 |
| 11 | 9.6460 | 7.2057 | 6.2167 | 5.6683 | 5.3160 | 5.0692 | 4.8861 | 4.7445 | 4.6315 |
| 12 | 9.3302 | 6.9266 | 5.9525 | 5.4120 | 5.0643 | 4.8206 | 4.6395 | 4.4994 | 4.3875 |
| 13 | 9.0738 | 6.7010 | 5.7394 | 5.2053 | 4.8616 | 4.6204 | 4.4410 | 4.3021 | 4.1911 |
| 14 | 8.8616 | 6.5149 | 5.5639 | 5.0354 | 4.6950 | 4.4558 | 4.2779 | 4.1399 | 4.0297 |
| 15 | 8.6831 | 6.3589 | 5.4170 | 4.8932 | 4.5556 | 4.3183 | 4.1415 | 4.0045 | 3.8948 |
| 16 | 8.5310 | 6.2262 | 5.2922 | 4.7726 | 4.4374 | 4.2016 | 4.0259 | 3.8896 | 3.7804 |
| 17 | 8.3997 | 6.1121 | 5.1850 | 4.6690 | 4.3359 | 4.1015 | 3.9267 | 3.7910 | 3.6822 |
| 18 | 8.2854 | 6.0129 | 5.0919 | 4.5790 | 4.2479 | 4.0146 | 3.8406 | 3.7054 | 3.5971 |
| 19 | 8.1849 | 5.9259 | 5.0103 | 4.5003 | 4.1708 | 3.9386 | 3.7653 | 3.6305 | 3.5225 |
| 20 | 8.0960 | 5.8489 | 4.9382 | 4.4307 | 4.1027 | 3.8714 | 3.6987 | 3.5644 | 3.4567 |
| 21 | 8.0166 | 5.7804 | 4.8740 | 4.3688 | 4.0421 | 3.8117 | 3.6396 | 3.5056 | 3.3981 |
| 22 | 7.9454 | 5.7190 | 4.8166 | 4.3134 | 3.9880 | 3.7583 | 3.5867 | 3.4530 | 3.3458 |
| 23 | 7.8811 | 5.6637 | 4.7649 | 4.2636 | 3.9392 | 3.7102 | 3.5390 | 3.4057 | 3.2986 |
| 24 | 7.8229 | 5.6136 | 4.7181 | 4.2184 | 3.8951 | 3.6667 | 3.4959 | 3.3629 | 3.2560 |
| 25 | 7.7698 | 5.5680 | 4.6755 | 4.1774 | 3.8550 | 3.6272 | 3.4568 | 3.3239 | 3.2172 |
| 26 | 7.7213 | 5.5263 | 4.6366 | 4.1400 | 3.8183 | 3.5911 | 3.4210 | 3.2884 | 3.1818 |
| 27 | 7.6767 | 5.4881 | 4.6009 | 4.1056 | 3.7848 | 3.5580 | 3.3882 | 3.2558 | 3.1494 |
| 28 | 7.6356 | 5.4529 | 4.5681 | 4.0740 | 3.7539 | 3.5276 | 3.3581 | 3.2259 | 3.1195 |
| 29 | 7.5977 | 5.4204 | 4.5378 | 4.0449 | 3.7254 | 3.4995 | 3.3303 | 3.1982 | 3.0920 |
| 30 | 7.5625 | 5.3903 | 4.5097 | 4.0179 | 3.6990 | 3.4735 | 3.3045 | 3.1726 | 3.0665 |
| 40 | 7.3141 | 5.1785 | 4.3126 | 3.8283 | 3.5138 | 3.2910 | 3.1238 | 2.9930 | 2.8876 |
| 60 | 7.0771 | 4.9774 | 4.1259 | 3.6490 | 3.3389 | 3.1187 | 2.9530 | 2.8233 | 2.7185 |
| 120 | 6.8509 | 4.7865 | 3.9491 | 3.4795 | 3.1735 | 2.9559 | 2.7918 | 2.6629 | 2.5586 |
| $\infty$ | 6.6349 | 4.6052 | 3.7816 | 3.3192 | 3.0173 | 2.8020 | 2.6393 | 2.5113 | 2.4073 |